THE FIRST $r+1$ COLUMNS OF THE CUBIC-DIMENSION SPECTRAL SEQUENCE OF A HYPERCUBE VIETORIS–RIPS COMPLEX

Abstract. Let $X^{n,r}$ be the Vietoris–Rips complex of the hypercube graph Q_n at scale r, filtered by cubic dimension, and let $E^{\bullet}_{\bullet,\bullet}$ be the resulting spectral sequence over a field of characteristic zero, all as in Galetto, Montaño and Wellner (arXiv:2606.20784). Problem 8.2 of that paper asks for a proof that $E^1_{p,0} \cong \mathcal{F}_p$, the p-th term of the cellular chain complex of the n-cube, and that $E^1_{p,q} = 0$ for $q \neq 0$, for all $0 \leq p \leq r$. We prove both, for every $n > r \geq 0$. Because $p \leq r$ the Rips condition is vacuous on the vertex set of a p-cell, so the p-th column is, exactly and not merely up to homotopy, the relative chain complex of a simplex modulo the union of its $2p$ half-cube faces; an affine marginal map identifies that pair $\mathcal{H}_p$-equivariantly with $(I^p, \partial I^p)$. For $r \leq 3$ both assertions are already proved in the source paper.

1. The problem

Let Q_n be the hypercube graph: vertex set $V(Q_n) = \{0,1\}^n$, with the Hamming distance d. For an integer $r \geq 0$ the *Vietoris–Rips complex* $X^{n,r}$ is the simplicial complex whose faces are the non-empty subsets $S \subseteq V(Q_n)$ with $\operatorname{diam}(S) \leq r$. The *cubic dimension* of such an S is

$$\operatorname{cdim}(S) = \#\{\, i \in [n] : x_i \neq y_i \text{ for some } x, y \in S \,\},$$

the number of coordinates in which S is not constant, and $X^{n,r}_p$ denotes the subcomplex of faces with cdim $\leq p$. This is an increasing filtration of $X^{n,r}$ by subcomplexes. Writing $C^{n,r}_{d,p}$ for the span, over a fixed field $\Bbbk$, of the faces of dimension d and cubic dimension *exactly* p, the associated homology spectral sequence has

$$E^0_{p,q} = C^{n,r}_{p+q,\,p}, \qquad E^1_{p,q} = H_{p+q}(C^{n,r}_{p+\bullet,\,p}).$$

The hyperoctahedral group $\mathcal{H}_n = (\mathbb{Z}/2)^n \rtimes S_n$ acts on Q_n by isometries, hence on everything above; all isomorphisms below are isomorphisms of $\mathcal{H}_n$-representations. Finally $\mathcal{F}_\bullet$ is the cellular chain complex of the n-cube $[0,1]^n$, so that $\mathcal{F}_p$ has as basis the p-cells of the cube and $\dim \mathcal{F}_p = \binom{n}{p} 2^{n-p}$.

All of the above is the setting of Galetto, Montaño and Wellner [1]; for other homological invariants of $X^{n,r}$ see [2]. Section 8 of [1], "Open problems", poses the following.

2020 *Mathematics Subject Classification.* 55N31, 05E18, 55T05, 05C10.

Key words and phrases. Vietoris–Rips complex, hypercube graph, cubic dimension, spectral sequence, hyperoctahedral group.

Problem 1.1 ([1], Problem 8.2)**.** Let $n > r \geq 0$. Show that for all integers p with $0 \leq p \leq r$ we have

(i) $E^1_{p,0} \cong \mathcal{F}_p$, where $\mathcal{F}_\bullet$ is the cellular chain complex of the n-cube; and
(ii) $E^1_{p,q} = 0$ for all $q \neq 0$.

The statement as posed quantifies explicitly only over p; the ambient quantifiers above are the ones we read off [1], whose main theorem is stated for $n > r$ and which fixes $\Bbbk$ to be of characteristic zero. The labels $\{\lambda; \mu\}$ for irreducible $\mathcal{H}_m$-representations are those of [1]; $\{0; 1^m\}$ is the determinant character of $\mathcal{H}_m$, i.e. the sign of the underlying permutation times the product of the signs.

2. The result

Theorem 2.1. *Let $n > r \geq 0$ and let $\Bbbk$ be a field of characteristic zero, as in* [1]*. Then for every p with $0 \leq p \leq r$:*

(i) $E^1_{p,0} \cong \mathcal{F}_p$ *as $\mathcal{H}_n$-representations; and*
(ii) $E^1_{p,q} = 0$ *for all $q \neq 0$.*

The characteristic hypothesis is inherited from the two results quoted from [1] in Propositions 2.2 and 2.3, and from the labelling $\{0; 1^p\}$; the local input we supply, Lemma 4.1, is characteristic-free, so over an arbitrary field the same argument still gives $E^1_{p,q} = 0$ for $q \neq 0$ and $\dim E^1_{p,0} = \dim \mathcal{F}_p$, with $\mathcal{H}_p$ acting by the determinant on each local summand.

For $r \leq 3$ both assertions of Problem 1.1 are already proved in [1] itself. We make no claim of priority beyond that, and none about literature we have not read. The proof below needs no computer; Section 5 describes the program that accompanies this note.

Two results quoted from [1]**, not reproved here.**

Proposition 2.2 ([1])**.** *For all p and q,*

$$E^1_{p,q} \;\cong\; \mathrm{Ind}^{\mathcal{H}_n}_{S_{n-p}\times\mathcal{H}_p}\big(\{n-p\} \otimes H_{p+q}(C^{p,r}_{p+\bullet,\,p})\big),$$

where $\{n-p\}$ is the trivial representation of S_{n-p}.

Proposition 2.3 ([1])**.** *For all i, $\mathcal{F}_i \cong \mathrm{Ind}^{\mathcal{H}_n}_{S_{n-i}\times\mathcal{H}_i}(\{n-i\} \otimes \{0; 1^i\})$, and $\{0; 1^i\}$ is one-dimensional. In particular $\dim \mathcal{F}_i = \binom{n}{i} 2^{n-i}$.*

These two are exactly the reduction we import. Induction is exact, so Proposition 2.2 says that the p-th column is determined by the case $n = p$ alone. Combining the two propositions:

Corollary 2.4. *Fix $r \geq 0$ and $0 \leq p \leq r$. If*

$$H_k(C^{p,r}_{p+\bullet,\,p}) = \begin{cases} \{0; 1^p\} & k = p, \\ 0 & k \neq p \end{cases} \qquad \textit{as } \mathcal{H}_p\textit{-representations,}$$

then assertions (i) *and* (ii) *of Problem 1.1 hold for that p and* every $n > r$.

Only this implication is used below. It is a *reduction*, not a proof: it converts Problem 1.1 into a statement about one local complex, and it supplies neither that complex's homology nor its character. Those are Sections 3 and 4.

3. The collapse

Fix p with $0 \le p \le r$ and put $n = p$; write $V = V(Q_p) = \{0,1\}^p$, so $|V| = 2^p$.

Lemma 3.1. *Since* $\operatorname{diam}(Q_p) = p \le r$*, the Rips condition is vacuous on V: every non-empty subset of V is a face of $X^{p,r}$. Hence*

$$\Delta := X^{p,r} = \text{the full simplex on the } 2^p \text{ vertices } V,$$

and a face $S \subseteq V$ has $\operatorname{cdim}(S) = p$ *if and only if S is* full-support*, i.e. S is non-constant in every coordinate. Therefore*

$$U := X^{p,r}_{p-1} = \bigcup_{i\in[p]} \ \bigcup_{\varepsilon\in\{0,1\}} \Delta(F_{i,\varepsilon}), \qquad F_{i,\varepsilon} := \{x \in V : x_i = \varepsilon\},$$

the union of the $2p$ half-cube simplices $\Delta(F_{i,\varepsilon})$ *(each a full simplex on 2^{p-1} vertices), and there is an equality of chain complexes of $\mathcal{H}_p$-representations*

$$C^{p,r}_{p+\bullet,\,p} \ = \ C_{p+\bullet}(\Delta, U). \tag{3.1}$$

Proof. Any two vertices of Q_p are at Hamming distance $\le p \le r$, so every non-empty $S \subseteq V$ satisfies $\operatorname{diam}(S) \le r$: this is the whole of the first assertion, and it is where the hypothesis $p \le r$ is used. Next, $\operatorname{cdim}(S) < p$ means exactly that S is constant in some coordinate i, i.e. $S \subseteq F_{i,\varepsilon}$ for some ε, which is the stated description of U; and $\operatorname{cdim}(S) = p$ means S is full-support. So the faces of Δ of cubic dimension exactly p are precisely the faces of Δ not in U, and $C^{p,r}_{d,p} = C_d(\Delta)/C_d(U) = C_d(\Delta, U)$ with the simplicial differential on both sides. Every step is $\mathcal{H}_p$-equivariant because $\mathcal{H}_p$ preserves the Hamming distance and hence cubic dimension. ☐

Equation (3.1) is an *equality*, not a homotopy equivalence, and it is the whole idea: the hypothesis $p \le r$ deletes all metric content, leaving an object with $2p$ pieces. At $p = 4$, for instance, Δ has $2^{16} - 1 = 65535$ faces and U has 1760, so $C^{4,4}_{4+\bullet,4}$ has 63775 generators; the next section obtains its homology from U, with no rank computation.

4. The local pair

Lemma 4.1 (the marginal map)**.** *There is an $\mathcal{H}_p$-equivariant homotopy equivalence of pairs*

$$(|\Delta|, |U|) \ \simeq \ (I^p, \partial I^p), \qquad I = [0,1],$$

where $\mathcal{H}_p$ acts on I^p as the affine symmetry group of the cube. Hence for all k

$$H_k(\Delta, U) \cong H_k(I^p, \partial I^p) = \begin{cases} \Bbbk & k = p, \\ 0 & k \neq p, \end{cases}$$

and $\mathcal{H}_p$ acts on $H_p(\Delta, U) \cong \Bbbk$ by the determinant of the linear part, i.e. by the character $\{0; 1^p\}$.

Proof. Realise $|\Delta|$ as the set of probability measures μ on $V = \{0,1\}^p$, the face of $|\Delta|$ containing μ in its interior being $\operatorname{supp}\mu$. Let

$$f : |\Delta| \longrightarrow I^p, \qquad f(\mu) = (\mathbb{E}_\mu x_1, \ldots, \mathbb{E}_\mu x_p),$$

an affine map, and let $s : I^p \to |\Delta|$ be the Bernoulli section $s(t) = \bigotimes_{i=1}^p \operatorname{Bern}(t_i)$, so that $f \circ s = \mathrm{id}$.

First, $f^{-1}(\partial I^p) = |U|$ *exactly*: $f(\mu)_i = \mu(\{x_i = 1\})$ lies in $\{0,1\}$ if and only if $\operatorname{supp}\mu \subseteq F_{i,0}$ or $\operatorname{supp}\mu \subseteq F_{i,1}$, i.e. if and only if $\operatorname{supp}\mu$ fails to be full-support in coordinate i; and by Lemma 3.1 that is exactly membership of U.

Second, the straight-line homotopy $H(\mu, u) = (1-u)\mu + u\, s(f(\mu))$ takes values in $|\Delta|$ (a convex combination of probability measures) and satisfies, because f is affine and $f \circ s = \mathrm{id}$,

$$f(H(\mu, u)) = (1-u)f(\mu) + u\, f(\mu) = f(\mu) \qquad \text{for all } u.$$

So f is constant along H; in particular H preserves $f^{-1}(\partial I^p) = |U|$ at every time. Thus H is a deformation retraction of the pair $(|\Delta|, |U|)$ onto $(s(I^p), s(\partial I^p))$, and s is a homeomorphism onto its image with inverse f.

Third, equivariance. For $g = (s, \sigma) \in \mathcal{H}_p$ acting on V by $(gx)_{\sigma(i)} = x_i + s_{\sigma(i)}$, one has $\mathbb{E}_{g\mu} x_{\sigma(i)} = \mathbb{E}_\mu x_i$ if $s_{\sigma(i)} = 0$ and $= 1 - \mathbb{E}_\mu x_i$ if $s_{\sigma(i)} = 1$. Hence f intertwines the $\mathcal{H}_p$-action on $|\Delta|$ with the affine action $t \mapsto (t_{\sigma^{-1}(i)} \text{ or } 1 - t_{\sigma^{-1}(i)})$ of $\mathcal{H}_p$ on I^p, which is the full affine symmetry group of the cube; s is equivariant for the same two actions, and H is then equivariant as well.

Finally $H_k(I^p, \partial I^p) = \widetilde{H}_{k-1}(S^{p-1})$ is $\Bbbk$ for $k = p$ and 0 otherwise (for $p = 0$ read $\partial I^0 = \emptyset$ and $H_0(\mathrm{pt}, \emptyset) = \Bbbk$), and an affine symmetry g of I^p acts on the fundamental class $[I^p, \partial I^p]$ by det of its linear part, which for $g = (s, \sigma)$ is $\operatorname{sgn}(\sigma)(-1)^{s_1 + \cdots + s_p}$. That is the determinant character of $\mathcal{H}_p$, i.e. $\{0; 1^p\}$ in the labelling of [1]. □

Proof of Theorem 2.1. By (3.1) and Lemma 4.1,

$$H_k(C^{p,r}_{p+\bullet,\, p}) = H_k(\Delta, U) = \begin{cases} \{0; 1^p\} & k = p, \\ 0 & k \neq p, \end{cases}$$

for every $0 \leq p \leq r$. Corollary 2.4 converts this into (i) and (ii) for every $n > r$. □

5. Scope, and the accompanying program

Three limits. First, the labelling $\{\lambda;\mu\}$ of $\mathcal{H}_m$-irreducibles is used as in [1], over a field of characteristic zero; Lemma 4.1 itself, and hence Theorem 2.1 read as "dim, and the determinant character", is characteristic-free. Second, the truncation $p \le r$ is essential to the argument: at $p = r+1$ the minimal full-support subsets of $V(Q_p)$ are the antipodal pairs, at distance $p > r$, so the Rips condition is no longer vacuous and Lemma 3.1 does not apply. Third, Theorem 2.1 is a statement about the terms $E^1_{p,q}$; we prove nothing here about the differential d^1, nor about the later pages.

Theorem 2.1 is proved above without a computer. A program `verify.py`, with its recorded output `verify.output.txt`, accompanies this note; the recorded run reports 144 checks, all passing. It rebuilds the local complexes $C^{p,r}_{p+\bullet,\,p}$ from the definitions of Section 1, checks (3.1) as an identity of basis sets for $p \le 4$, and confirms the conclusion of Lemma 4.1 — one class, in degree p, on which $\mathcal{H}_p$ acts by the determinant — for $p \le 4$, computing the homology of the 63775-generator complex $C^{4,4}_{4+\bullet,4}$ both through U and directly by exact elimination on its fifteen boundary matrices. It also reproduces the dimensions [1] publishes for its own scale-2 and scale-3 computations. It is Python 3.9+, standard library only; all arithmetic is exact integer arithmetic, and ranks are taken modulo a prime, which bounds the characteristic-zero Betti numbers from above, the field-independent alternating sum then pinning them.

The program reproves neither Proposition 2.2 nor Proposition 2.3, does not model the homotopy of Lemma 4.1, and enumerates no cell with $p = r \ge 5$, which would require 2^{32} subsets; its output lists each check it runs and each thing it does not cover, and some of those checks concern objects the argument above does not use. The proof in Sections 3 and 4 is uniform in n, r and p and does not depend on any computation.

References

[1] F. Galetto, J. Montaño, and Z. Wellner, *Homology of Vietoris–Rips complexes of hypercube graphs via group actions*, arXiv:2606.20784v1, 2026. arXiv:2606.20784.

[2] M. Bendersky, N. Elia, and J. Grbić, *Cohomological properties of the Vietoris–Rips complex of a hypercube graph*, arXiv:2605.00705v2, 2026. arXiv:2605.00705.